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INCH-POUND
MIL-HDBK-1884
14 March 1998

DEPARTMENT OF DEFENSE
HANDBOOK

COATING, PLASMA SPRAY DEPOSITION



AMSC N/A AREA MFFP
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FOREWORD

1. This Military Standard is approved for use by U.S. Army Tank- Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Purpose	1
1.2	Scope	1
2.	APPLICABLE DOCUMENTS	2
2.1	Government Documents	2
2.1.1	Specifications, standards, and handbooks	2
2.1.2	Other Government documents, drawings, and publications	2
2.2	Non-Government publications	2
2.3	Order of precedence	4
3.	DEFINITIONS	5
3.1	Rotating part	5
3.2	Non-rotating (stationary part)	5
3.3	Plasma lot	5
3.4	Powder lot	5
3.5	Simulated part	5
4.	GENERAL REQUIREMENTS	6
4.1	Equipment	6
4.1.1	Plasma spray gun	6
4.1.2	Automatic plasma spray equipment	6
4.1.3	Cup test machine	6
4.2	Materials	6
4.2.1	Plasma gas	6
4.2.1.1	Restrictions	6
4.2.2	Coating powder	6
4.2.3	Bonding adhesive	6
4.3	Required procedures and operations	6
4.3.1	Surface machining	6
4.3.2	Heat treatment	7
4.3.3	Surface treatment	7
4.3.4	Surface preparation	7
4.3.4.1	Cleaning	7
4.3.4.2	Handling and storage	7
4.3.4.3	Overspray protection	7
4.3.4.3.1	Tolerances	7
4.3.5	Abrasive blast	7

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CONTENTS

Paragraph		Page
4.3.6	Preheating	7
4.3.7	Coating deposition	8
4.3.7.1	Process control	8
4.3.8	Substrate cooling	8
4.3.9	Sealing	8
4.3.10	Identification	8
4.4	Operator certification	8
4.4.1	Plasma spray operator	8
4.4.1.1	Continuance of qualifications	9
5.	DETAILED REQUIREMENTS	10
5.1	General	10
5.1.1	Qualifications	10
5.1.1.1	Qualification sample	10
5.1.1.2	Requalification	10
5.1.2	Test samples	10
5.1.2.1	Cup test panels	10
5.1.2.2	Bond strength specimens	10
5.1.2.3	Metallographic test specimens	11
5.1.3	Process approvals	11
5.1.3.1	Processing data	11
5.1.3.2	Sample part	11
5.1.3.3	Reworks	11
5.1.3.3.1	Stripping	11
5.1.3.4	Metallographic test sample	11
5.1.4	Process certification	11
5.2	Test methods	12
5.2.1	Visual	12
5.2.2	Cup test	12
5.2.3	Bond strength test	12
5.2.4	Micro examination	13
5.2.4.1	Oxides and porosity	13
6.	NOTES	14
6.1	Intended use	14
6.2	Issue of DoDISS	14
6.3	Supersession data	14
6.4	Safety	14
6.5	Cup test machine	14

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Check the source to verify that this is the current version before use.

CONTENTS

Paragraph		Page
6.6	Subject term (key word) listing	14
6.7	Changes from previous issue	14

Table		Page
1.	Typical plasma spray control test	15
2.	Materials for test specimens	16

Figure		Page
1	Bond strength test specimen	17

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1. SCOPE

1.1 Purpose. The procedures covered by this standard are intended to ensure that plasma spray coating operations, either manual or automated, meet prescribed requirements. This process can be used on parts requiring protection from wear, heat, corrosion (with sealer), abrasion and to restore dimensionally discrepant parts.

1.2 Scope. This standard covers the requirements for thermal coating by means of plasma spray deposition.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-P-80109 - Plasma Spray Systems, Powder, Guns, and Accessories.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Printing Service Office, Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

U.S. DEPARTMENT OF LABOR

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

OSHA - Safety and Health Standards (29 CFR 1910) General Industry.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 2750	- Pyrometry.
AMS 4026	- Aluminum Alloy, Sheet and Plate 1.0 Mg-0.6 Si-0.28 Cu-0.20 Cr- (6061; -T4) Sheet, -T451 Plate Solution Heat Treated and Naturally Aged.
AMS 4117	- Aluminum Alloy Bars, Rods, and Wire Rolled or Cold Finished and Flash Welded Rings 1.0 Mg-0.6 Si-0.28 Cu-0.20 Cr (6060-T4).
AMS 4901	- Titanium Sheet, Strip, and Plate Annealed, 70,000 psi yield Strength.
AMS 4928	- Titanium Alloy Bars, Wire, Forgings, and Rings 6A1-4V Annealed.
AMS 6350	- Steel Sheet, Strip, and Plate 0.95 Cr-0.20 Mo (0.28-0.33 C).
AMS 6382	- Steel Bars, Forgings, and Rings 0.95 Cr-0.20 Mo (0.38-0.43 C).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Avenue, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E3	- Standard Methods of Preparation of Metallographic Specimens.
ASTM E407	- Standard Test Methods for Microetching Metals and Alloys.
ASTM E643	- Ball Punch Deformation of Metallic Sheet Material, Test Method for.

Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS Welding Handbook, Volume 3, Seventh Edition, Chapter 12 “Thermal Spraying”.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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AMERICAN SOCIETY FOR METALS (ASM)

ASM Handbook, Volume 2, Eighth Edition, “Metal Spraying”.

(Application for copies should be addressed to the American Society for Metals, Metals Park, OH 44073.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Rotating part. Rotating parts are defined as those turning on an axis such as bearing plates, discs, hubs, shafts, etc.

3.2 Non-rotating (stationary) part. Non-rotating parts are defined as nonmoving or fixed parts.

3.3 Plasma lot. A plasma lot shall consist of all the parts (with the same part number) plasma sprayed by the same operator per shift using the same process and equipment and the same powder lot.

3.4 Powder lot. A powder lot shall be all the powder of a specified type with the same batch lot number to be used for coating.

3.5 Simulated part. A simulated part is a test piece or section with a similar surface configuration to the part it represents. The simulated part shall be approved by the acquisitioning authority materials laboratory.

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4. GENERAL REQUIREMENTS

4.1 Equipment.

4.1.1 Plasma spray gun. The plasma spray gun shall be a specially constructed torch that disassociates and ionizes a suitable plasma gas when a high voltage arc is struck between an anode and cathode. The gun shall provide for the injection of powder materials into a high velocity flame and high temperature arc wherein the powder materials can be heated to plasticity and propelled toward the workpiece to be coated as specified in MIL-P-80109.

4.1.2 Automatic spray equipment. When the plasma spray process is done by automated equipment, the preheating and process temperature checks may be waived providing prior approval is obtained in writing from the acquisitioning authority materials laboratory.

4.1.3 Cup test machine. A cup test machine equipped with dies as specified in ASTM E643 to form a spherical cup shape shall be used for determining bond adhesion (see 6.2). This machine shall be utilized to conduct cup tests on coated panels prepared in accordance with 5.1.2.1. These panels shall meet requirements after testing as specified in 5.2.2.

4.2 Material.

4.2.1 Plasma gas. Unless otherwise specified, the plasma gas shall be any of the following:

a. Nitrogen (N_2).
b. Mixture of nitrogen and hydrogen ($N_2 + H_2$) (see 4.2.1.1).
c. Argon (Ar).
d. Mixture of argon and hydrogen ($Ar + H_2$).
e. Mixture of argon and helium (Ar + He).

NOTE: Nitrogen shall not be used for spraying titanium parts.

4.2.1.1 Restrictions. Plasma gas consisting of nitrogen or nitrogen mixtures shall only be used when approved by the acquisitioning authority materials laboratory.

4.2.2 Coating powder. The powder for coating shall be as specified on the engineering drawing. It shall be dry, free flowing, and thoroughly blended.

4.2.3 Bonding adhesive. The adhesive bonding material used for the bond strength test shall be capable of producing a tensile bond strength greater than the strength of the coating material to be tested.

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4.3 Required procedures and operations.

4.3.1 Surface machining. When required, surfaces may be machined undersize or holes oversize to allow for finish thickness of the coating.

4.3.2 Heat treatment. When required, the heat treatment of parts shall be performed before coating.

4.3.3 Surface treatments. Unless otherwise specified, all surface treatments (e.g., black oxide, phosphate, anodize, etc.) as specified on the engineering drawing, shall be done after plasma spraying.

4.3.4 Surface preparation.

4.3.4.1 Cleaning. Part surfaces shall be clean and free from moisture, oil, grease, dirt, scale, paint or other foreign matter detrimental to plasma spraying. Parts shall be thoroughly cleaned by vapor degreasing or washing in petroleum solvents. Other solvents may be used providing the solvents leave no residue on the work surface. The chloride content for all liquids used for cleaning or rinsing titanium and titanium alloy parts shall not exceed 200 parts per million (ppm).

4.3.4.2 Handling and storage:

a. Handling. All surfaces requiring plasma spraying shall be handled with clean lint-free gloves, tongs, or other means that will preclude surface contamination.
b. Storage. If a delay in spraying occurs beyond 2 hours but less than 20 hours, special measures shall be employed to protect the surface to be coated from dust, dirt, moisture, etc. Protection shall be in the form of covering or inserting the parts in clean plastic bags. Alternative methods may be to store parts overnight in a moderate temperature oven maintained at 200 $\pm$ 5 degrees Fahrenheit (°F) or in a vacuum chamber under low pressure. Should the delay in spraying, after proper surface preparation, exceed 20 hours the parts shall be reprocessed in accordance with 4.3.4.1.

4.3.4.3 Overspray protection. Areas adjacent to the area to be plasma spray coated shall be suitably protected from overspray by masking or shielding.

4.3.4.3.1 Tolerances. Unless otherwise specified on the engineering drawing, a tolerance of -0 and +1/8 inch shall be allowed on the boundaries of areas designated to be coated.

4.3.5 Abrasive blast. All surfaces designated for plasma coating shall be roughened by abrasive blasting with the coarsest nonmetallic abrasive practical for the specified base material.

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4.3.6 Preheating. Unless otherwise specified (see 4.1.2), surfaces shall be preheated by a suitable and controllable source. The following shall apply:

a. Preheating is performed to remove moisture and minimize the thermal shock encountered during deposition.
b. Temperature of the part, during the preheating and coating application, shall be controlled to prevent discoloration, oxidation, distortion and other conditions detrimental to the coating or substrate.
c. Temperature of the part after preheat and prior to spraying shall be measured using the appropriate pyrometric devices in accordance with AMS 2750.

CAUTION: Special care must be taken to avoid overheating the lower temperature melting nonferrous alloys such as aluminum and magnesium.

4.3.7 Coating deposition. The coating material shall be deposited on the designated surfaces to a sufficient thickness to provide, after subsequent operations, a finished coating thickness which will meet the engineering drawing dimensions.

4.3.7.1 Process control. Process control procedures shall be checked during plasma spraying to assure the specified operating parameters on the control sheet are maintained (see table I).

4.3.8 Substrate cooling. Because the plasma flame imparts a large amount of heat to the workplace, rapid dissipation of heat at the point of flame to work contact is necessary to prevent overheating and oxidation of 400°F during spraying. Substrate cooling is usually achieved by an external blast of gas (air, nitrogen, carbon dioxide, or argon) impinging on the substrate either/or on both sides of the torch flame.

4.3.9 Sealing (post treatment). When specified, coated parts shall be sealed to extend their service life or prevent corrosion of the substrate. The appropriate sealer may be applied to fill only subsurface pores in the deposit or subsurface irregularities. The latter technique will provide a smooth coating to resist industrial atmospheres. There are various sealers designed for specific requirements; more specific information may be obtained from either the ASM Handbook Volume 2, Eighth Edition, or the AWS Handbook, Volume 3, Seventh Edition.

4.3.10 Identification. Each coated part shall be identified by applying the symbol S as a prefix to the serial number (e.g, S S/N 1234) or as a prefix to the Federal manufacturer code on nonserialized parts. The prefix shall be applied in the same manner as specified on the engineering drawing for the part number. In the event that available space precludes application of the symbol as a prefix, the symbol shall be applied as closely as possible to the serial number or Federal manufactures code. In no case shall the symbol be applied as a suffix to the serial number or Federal manufactures code.

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4.4 Operator certification.

4.4.1 Plasma spray operator. The plasma spray operator shall demonstrate to the acquisitioning authority and supplier metallurgical engineer suitable proficiency in the operation of the plasma spray equipment and in the performance of other related items by successfully preparing and coating a set of test panels and specimens as described in 5.1.2 through 5.1.2.3. In the event an operator fails the test, the operator shall be permitted to submit two more samples of the test panels or specimens which failed to pass the specified test. If any of the second set fail to conform to the specified acceptance requirements, the operator shall be disqualified. If re-examination is still desired after failing to pass the second set the operator shall be required to submit a complete series of qualification test panels and specimens as specified in 5.1.2 through 5.1.2.3. In addition, the operator shall be required to submit one additional set of the test panels or specimens which resulted in the initial disqualification. A plasma spray operator of semi-automatic or automatic equipment shall meet all the requirements of a manual spray operator.

4.4.1.1 Continuance of qualification. The qualification status of a plasma spray operator may be maintained by a continuous record of satisfactory proficiency or by annually successfully preparing and coating a set of test panels and specimens as described in 5.1.2.1 through 5.1.2.3. Requalification of a plasma spray operator shall be required when there is evidence of a lack of proficiency.

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5. DETAILED REQUIREMENTS

5.1 General.

5.1.1 Qualification. To ensure supplier capability to consistently deliver a satisfactory product, the supplier facilities and general procedures shall be qualified before parts for production are supplied unless such qualification is waived by the acquisitioning authority materials laboratory.

5.1.1.1 Qualification samples. The supplier shall submit test samples as specified in 5.1.2, together with the sample parts required for examination by the acquisitioning authority materials laboratory. The supplier shall plasma spray the coating and base metal combination as specified by the acquisitioning authority. Sample parts submitted shall be prepared in accordance with 5.1.3.2.

5.1.1.2 Requalification. In the event a supplier has not supplied plasma sprayed parts or test specimens for over 1 year, the supplier shall be required to requalify in a manner specified by the acquisitioning authority materials laboratory.

5.1.2 Test samples. A minimum of two panels for cup test, one panel for microexamination and two bond strength specimens shall be plasma sprayed. The cup, and microscopic examination panels and bond specimens shall be fabricated from the materials specified in table II.

5.1.2.1 Cup test panels. Cup test panels shall be prepared as follows:

a. Size: 3.00 x 1.75 x 0.050 inches.
b. Material: Alloy specified in table II.
c. Coating: Specified by acquisition authority.
d. Coating thickness: 4 to 6 mils or 2 to 4 mils when specified by the acquisitioning authority.

5.1.2.2 Bond strength specimens. Bond strength specimens shall be prepared as follows:

a. Size. Two specimens to the dimensions as specified in figure 1. Other specimen configurations may be used providing written permission is obtained from the acquisitioning authority material laboratory.
b. Material. One specimen of the selected alloy as specified in 5.1.2 and one specimen of any suitable material for the uncoated end of the test specimen.
c. Coating. Only the specimen of the similar alloy as the parts being coated is to be coated using the coating material as specified by the acquisitioning authority.

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NOTE: The uncoated specimen shall be suitably identified (e.g., slotted along the outside diameter).

d. Coating thickness. 9 to 12 mils.

5.1.2.3 Metallographic test specimens. The same material and size requirements as the cup test panel shall apply to the microscopic test specimens (see 5.1.2.1). Coating thickness shall be 9 to 12 mils unless otherwise specified by the procuring activity. Specimen configuration other than the shape specified shall require the approval of the procuring activity.

5.1.3 Process approvals.

5.1.3.1 Processing data. Operation sheets covering the coating process shall be established for each part number by the plasma spray source. Table I shows a typical operation sheet, the format is optional provided all pertinent information is shown. Processing procedures shall not be dependent upon part function (rotating versus nonrotating) or the critical nature when in service. The engineering requirements as specified on the drawing shall prevail during any acceptance testing.

5.1.3.2 Sample part. A sample part, or a simulated part shall be processed in accordance with the procedure established in 5.1.3.1 and submitted to the acquisitioning authority for approval prior to production parts being plasma sprayed.

5.1.3.3 Rework. Rework of discrepant parts with damaged or imperfect coatings, shall be reworked by a qualified source approved by the acquisitioning authority. The number of reworks will also be an acquisitioning authority decision based on degree discrepant and method of repair.

5.1.3.3.1 Stripping. Reworking to completely remove the defective coating is permissible by mechanical methods. Nonmechanical (chemical) stripping shall be subject to approval of the acquisitioning authority.

5.1.3.4 Metallographic test sample. Metallographic test samples shall be prepared in accordance with 5.1.2.1 and plasma sprayed along with the first production lot of parts and each subsequent plasma lot (see 3.3) and submitted to the acquisitioning authority with the parts. When approved by the acquisitioning authority materials laboratory, the plasma spray supplier may submit microphotographs of the coating in lieu of the test samples.

5.1.4 Process certification. Unless otherwise specified, the supplier shall furnish three copies of a certificate stating the following information:

a. Purchase order number.
b. Part number and revision letter.

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c. Serial number of parts coated (when applicable).
d. Plasma lot numbers.
e. Quantity and serial numbers of rejected parts stripped and recoated.
f. Test results per 5.2.1 through 5.2.4.
g. Statement of conformance to this standards latest revision.

5.2 Test methods.

5.2.1 Visual. All parts and assemblies (before and after machining), test panels, and test specimens, shall be visually inspected to verify that plasma coating is adherent to the substrate material and has a uniform continuous surface free from spalling, chipping, flaking, cracking, and other objectionable imperfections.

5.2.2 Cup test. A test panel (see 5.1.2.1) shall be cup tested to verify conformance to the applicable coating material specification requirements as follow:

a Draw the panel on a testing machine equipped with dies (see 4.1.3) to form a cup shape as specified in ASTM E643. The test panel shall be drawn at a prescribed rate with the coated side of the panel being opposite the punch.
b. Draw the panel to the depth specified in the applicable coating material specification.
c. Examine coated surface of panel for indications of coating failure. Cracking shall be acceptable provided there is no evidence of flaking or spalling. Flaking or spalling of the coating shall be rejected.

NOTE: Other devices may be used for cup testing provided equivalent results are obtained and it is approved by the acquisitioning authority materials laboratory.

5.2.3 Bond strength test. Two bond strength test specimens (see 5.1.2.2) shall be bonded together to form a single test specimen and the test specimen shall be bond strength tested to verify conformance to the applicable coating material requirements as follow:

a. Remove masking materials.
b. Verify that bonding end of the uncoated section of the test specimen has been cleaned, dried, and abrasive-blasted.
c. Apply an adhesive (see 4.2.3) to the bonding ends of the coated and uncoated sections of the test specimen.
d. Accurately align and join the ends of the two specimens.

NOTE: A fixture may be used to facilitate bonding of the test specimen.

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e. Cure the adhesive in a hot air circulating oven at the temperature recommended by the manufacturer.
f. After bonding, dress the edge of the coating flush with the outside diameter of the test specimen.

NOTE: An abrasive disc or wheel may be used for this purpose but care shall be taken that the disc or wheel moves parallel to the centerline of the test specimen. Care shall be taken during dressing to ensure the temperature does not increase to affect the bond adhesion.

g. Test the bond specimen with a standard laboratory tensile tester equipped with universal joint grippers for each end of the joined specimens. Set the no-load cross-head speed at approximately 0.050 inch per minute.

5.2.4 Micro examination. A microscopic examination of a specimen cut normal to the coating surface shall not reveal any cracks, excessive or massive oxides or porosity when examined at an appropriate magnification in the unetched condition in accordance with the procedures outlined in ASTM E3 and ASTM E407.

5.2.4.1 Oxides and porosity. Excessive porosity shall be defined as not greater than 30 percent when viewed at 100x after etching. Excessive or massive oxides shall be defined as not greater than 3 percent for coatings where oxygen is not an integral part of the material. Examination of the part for oxides shall be at 100x after etching.

NOTES: The metallurgical specimen initially approved for a particular plasma spray application shall be the minimum acceptance standard for that plasma spray application.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This military standard is intended to ensure plasma spray coating operations on parts for military vehicles meet prescribed requirements.

6.2 Issue of DoDISS. When this standard is used in acquisition, the applicable issue of the DoDISS must be cited in the solicitation (see 2.1.1 and 2.2).

6.3 Supersession data. This standard supersedes MIL-STD-1884(AT) dated 29 July 1985 and Textron Lycoming, Stratford, Connecticut, Specification No. P6405F entitled "Coating, Plasma Spray Deposition", dated 4 April 1991.

6.4 Safety. Use of plasma spray equipment involves exposure to certain safety hazards specified in OSHA Safety and Health Standards (29 CFR 1910), General Industry.

6.5 Cup test machine. The base machine to be used for this test shall be the industry accepted model available from Detroit Testing Machine Co., Detroit, Michigan or an approved equivalent (see 4.1.3).

6.6 Subject term (key word) listing.

Coating
Plasma

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Typical plasma spray control sheet.
Sheet __ of __.

VENDOR _____ VENDOR PROCESS # _____
PURCHASE ORDER NUMBER _____
PART NUMBER _____ S/N _____
AREA TO BE COATED _____
PART MATERIAL _____
GUN TYPE _____ GUN MODEL _____ NOZZLE _____

PART PREPARATION
METHOD OF CLEANING _____
MASKING INFORMATION _____
GRIT TYPE AND SIZE _____
GRIT BLAST PRESSURE (PSI) _____ ± Suction _____ Pressure _____
NOZZLE TO WORK DISTANCE _____

ARC GAS
GAS (1) PRIMARY _____
GAS (2) SECONDARY _____

Regulator (1)	PSI _____ ±
Regulator (2)	PSI _____ ±
Console (1)	PSI _____ ±
Console (2)	PSI _____ ±
Console Flow (1)	C.F.H. _____ ±
Console Flow (2)	C.F.H. _____ ±

POWER
Voltage DC Operating _____ ±
Amperes DC Operating _____ ±

COATING MATERIAL
Material Specification _____ Other _____
Material Lot # _____

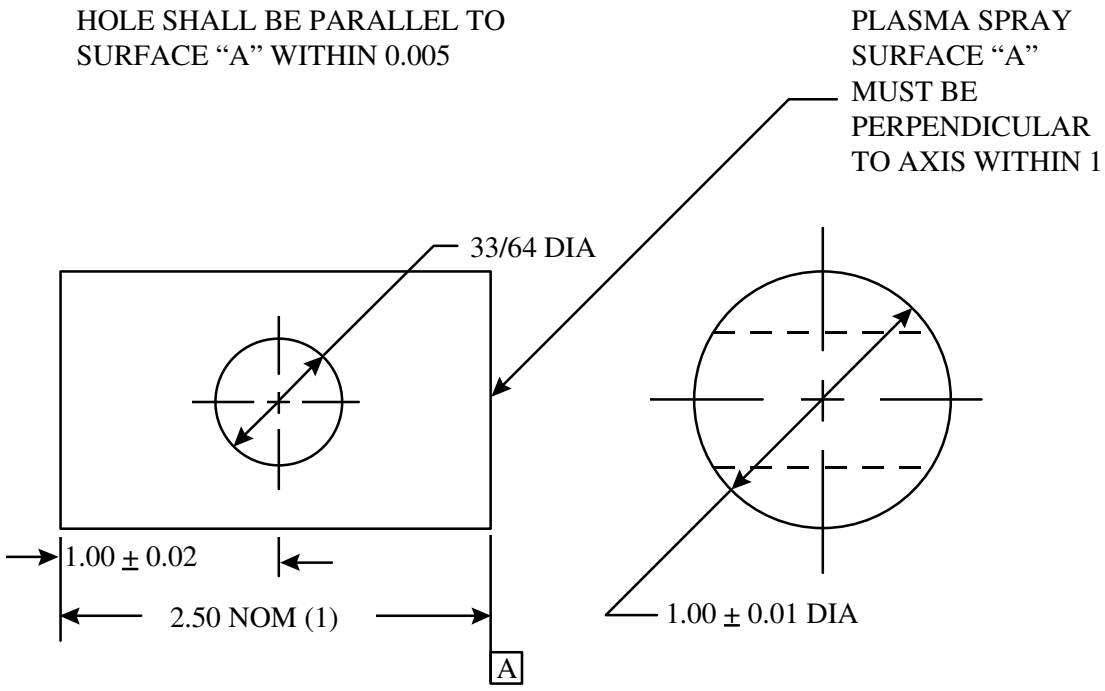
COATING DATA
Elapsed Time Between Surface Prep and Spraying _____
Powder Feed Rate (LB/HR) _____
Powder Port _____ Meter Wheel _____
Gun to Work Distance _____ Carrier Gas Flow _____
Minimum Gun to Work Angle _____
Part RPM _____ ± _____ Surface Feed/Min _____ ±
Coating Thickness as Sprayed _____
Preheat Temp _____ ± Method of Preheat _____
Max Part Temp During Spraying _____
Number of Passes Per Cycle _____
Cool Timer Per Cycle _____ ±
Method of Cooling Air _____ Gas _____ Static _____
Total Number of Cycles (Cool Spray) _____ ±
CERTIFICATION # _____
APPROVAL _____

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TABLE II. Materials for test specimens.

Part base material	Bond specimens	Cup and micro specimens
Steel, nickel, cobalt	AMS 6382	AMS 6350
Titanium or titanium alloys	AMS 4928	AMS 4901
Aluminum or aluminum alloys	AMS 4117	AMS 4026
Magnesium or magnesium alloys	As specified by acquisitioning authority.	

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NOTE:

(1) Actual specimen length may be varied to suit test facilities.

(2) All dimensions are given in inches.

FIGURE 1. Bond strength test specimen.

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Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project MFFP-A496)

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